

STRAIN GAUGES

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Strain gauges is a thin, wafer-like device that can be attached (bonded) to a variety of materials to measure applied strain.

Metallic strain gauges are manufactured from small diameter resistance wire such as constantan or etched from ^{thin} sheet foil sheets. The resistance of the wire or metal foil changes with length as the material to which the gauge is attached undergoes tension or compression. This change in resistance is proportional to the applied strain and is measured with specially adapted Wheatstone bridge.

OPERATING PRINCIPLE OF RESISTANCE STRAIN GAUGE

The working of strain gauge is based on the fact that when stress is applied on the metal conductor its resistance changes owing to change in length and cross-sectional area of the conductor as shown in figure below. Resistance of conductor under stress is also changed due to change in resistivity this property is called the piezo-resistive effect. That is why strain gauges are called the piezo-resistive strain gauges.

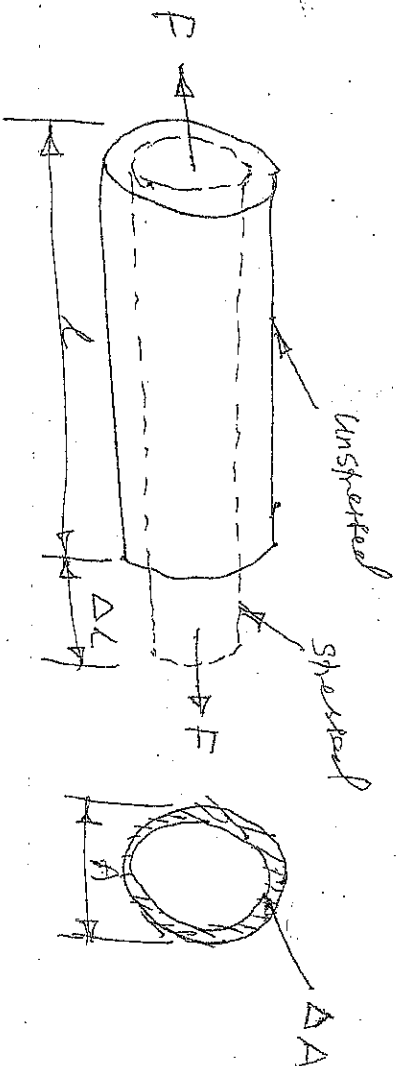


Figure Conductor under axial tension

If a conductor of length L , area of cross-section A is subjected to axial tension, the resistance will change because

of change in length, cross-sectional area and resistivity of the material.

Resistance of an unstrained conductor is given by ~~equation~~ ^{equation}

$$R = \frac{\rho l}{A} \quad \dots (1)$$

Let under strained conditions resistance of conductor be changed by ΔR because of change in length by Δl , cross-sectional area by ΔA and resistivity by $\Delta \rho$.

The sensitivity of a strain gauge is described in terms of a characteristic called the gauge factor K , defined as the unit change in resistance per unit change in length.

The gauge factor can be written as

$$G = \frac{\Delta R/R}{\Delta l/l} = 1 + 2\mu + \frac{\Delta \rho/\rho}{\Delta l/l} \quad \dots (2)$$

Change in resistance due to change in length Change in resistance due to change in area Change in resistivity due to piezo-resistive effect

Thus the strain $\Delta l/l$ can be measured in terms of gauge factor G and $\Delta R/R$.

The strain $\Delta l/l = \epsilon$ and is usually measured in microm - strain ($\pm \mu\text{m/m}$).

The term $\frac{\Delta \rho/\rho}{\Delta l/l}$ can be expressed as πE , where π is

called the longitudinal piezo-resistance coefficient and E is called the Young's modulus of elasticity.

The material property π may be either the or pie. Poisson's ratio for most of the metals lies in the range of 0 and 0.5.

From equation (2) it can be seen that the change in resistance of a ~~material~~ ^{metal} wire due to strain is caused by

1. - ~~change in length~~ ^{change in length} ~~the change in resistance~~ ^{the change in resistance}

represented by the factor $(1 + 2\mu)$ and change in ② resistivity $\Delta\rho$. Poisson's ratio μ is almost constant and is equal to 0.3, the strain sensitivity or gauge factor G is 1.6 when considering only the dimensional change aspect.

Thus for an elastic material the Poisson's ratio μ is defined as the ratio of lateral strain to longitudinal strain. i.e.

$$\mu = \frac{\text{lateral strain}}{\text{longitudinal strain}}$$

Stress is directly proportional to strain, within elastic limit of the material, and the ratio of stress to strain is called Young's Modulus of elasticity of material, E .

$$1 \text{ l } E = \frac{\text{stress}}{\text{strain}} = \frac{\rho}{\epsilon} \quad \text{kg/m}^2$$

Where ρ is axial stress in kg/m^2 and ϵ axial strain.

The gauge factor of the device $G = \frac{\Delta R/R}{\epsilon}$

Example 1

A resistance wire strain gauge uses a soft iron wire of small diameter. The gauge factor is +5.2. Neglect the piezo-resistive effect, Calculate the Poisson's ratio.

Soln:-

$$\text{From } G = 1 + 2\mu + \frac{\Delta\rho/\rho}{\Delta l/l}$$

Since piezo-resistive effect is negligible, the last term $\frac{\Delta\rho/\rho}{\Delta l/l}$ becomes negligible and the equation for gauge factor becomes $G = 1 + 2\mu$

or Poisson's ratio, $\mu = \frac{G-1}{2} = \frac{5.2-1}{2} = 2.1$

Example 2

Calculate the gauge factor of a strain gauge if a 1.5 mm diameter conductor that is 24 mm long changes length by 1 mm and diameter by 0.02 mm under a compression force.

Soln:-

$$\text{Poisson's ratio, } \mu = \frac{\text{Lateral Strain}}{\text{Longitudinal Strain}} = \frac{\Delta d/d}{\Delta L/L}$$

$$= \frac{0.02/1.5}{1/24} = \frac{0.02 \times 24}{1 \times 1.5} = 0.32$$

$$\text{Gauge factor } G = 1 + 2\mu = 1 + 2 \times 0.32 = 1.64$$

Example 3

A strain gauge with gauge factor of 2 is fastened to a metallic member subjected to a stress of 1000 kg/cm^2 . The modulus of elasticity of the metal is $2 \times 10^6 \text{ kg/cm}^2$. Calculate the percentage change in resistance of the strain gauge.

(i) The Poisson's ratio of the gauge.

Soln:-

$$\text{(i) Strain, } \epsilon = \frac{f}{E} = \frac{1000}{2 \times 10^6} = 0.0005$$

$$\Delta R/R = \epsilon \times G = 0.0005 \times 2 = 0.001$$

Percentage change in resistance = $\Delta R/R \times 100 = 0.001 \times 100 = 0.1\%$

(ii) Since $G = 1 + 2\mu$ where μ is Poisson's ratio

$$\text{Then } \mu = \frac{G-1}{2} = \frac{2-1}{2} = 0.5$$

Based on mounting, strain gauges are classified as:
Bonded strain gauges and unbonded strain gauges.

1. Bonded strain gauge - Bonded strain gauges are attached to the elastic element surface. In bonded resistance wire strain gauges resistance element is bonded to the base which may be a thin sheet of paper or a thin sheet of bakelite or teflon. The bonded strain gauge is connected to the Wheatstone bridge circuit. The wire is covered on top with a thin sheet of material in order to protect it from any mechanical damage. The spreading of wires allows a uniform distribution of stress over the grid. Two connecting leads are soldered or welded to the ends of the grid. The bonded strain gauge is connected with a special adhesive to the structure whose tensile or compressive strain is to be measured.

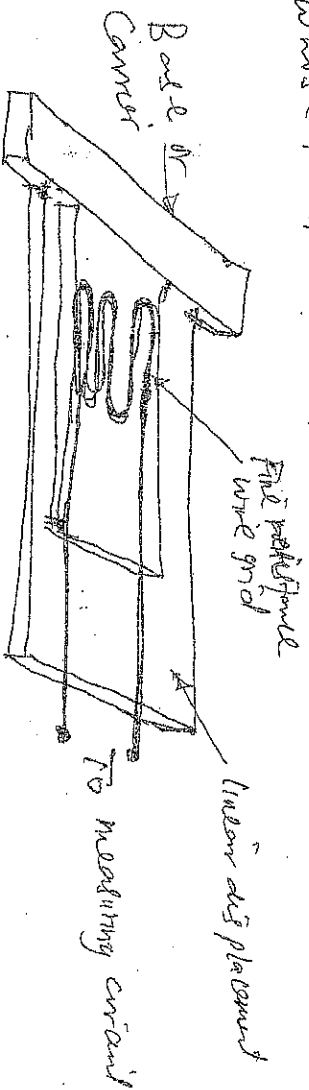


Figure Bonded strain gauge

2. Unbonded strain gauge - In such type of gauge, resistance wire is not bonded on a base, instead of that strain is directly transferred to the resistance wire and so a smaller force is required for changing the wire length. In such a strain gauge the resistance wire is about 25 micron diameter and stretched between a supporting frame and an armature that is supported in the center of the frame as shown in figure.

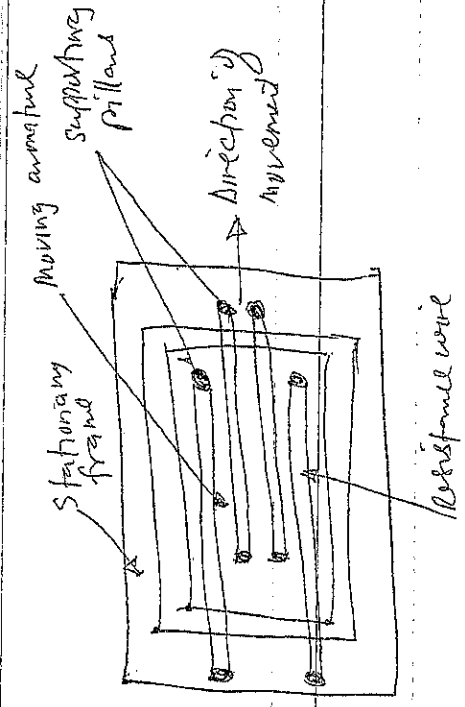
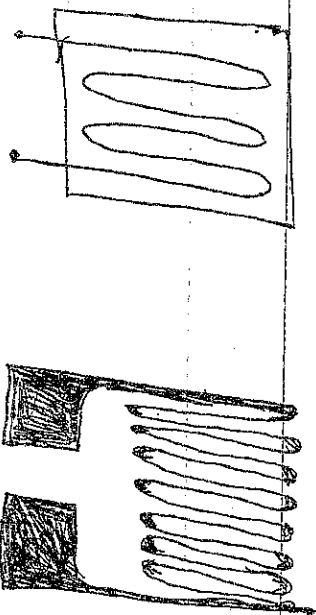


Figure Unbonded Strain gauge.

WIRE STRAIN GAUGE

(4)

The resistance element is in the form of wire foil or film of the material.



(a) wire

(b) metal foil

Figure strain gauge element.

The strain gauge - The resistance element is in the form of the foil or film of the material.

Method for - The strain is detected by using a metal foil as shown in figure (b). The metal foil is pasted on one side of plastic. The leads are soldered to metal foil for connecting the Wheatstone bridge. Metal foil strain gauges exhibit a hysteresis gauge factor than wire foil strain gauges. The metals and alloys used for the foil or wire are - Constantan (Cu-Ni), Nichrome (Ni-Cr), Incoelastic (Fe-Ni-Cr-Mn-Mo), Platinum-Tungsten (Pt-W).

TYPES OF ELECTRICAL RESISTANCE STRAIN GAUGE

A strain gauge is a sensor whose resistance varies with an applied force. The strain gauge converts the force, pressure, tension, and weight into electrical quantity which can be measured.

The strain gauge

ROSETTE STRAIN GAUGES

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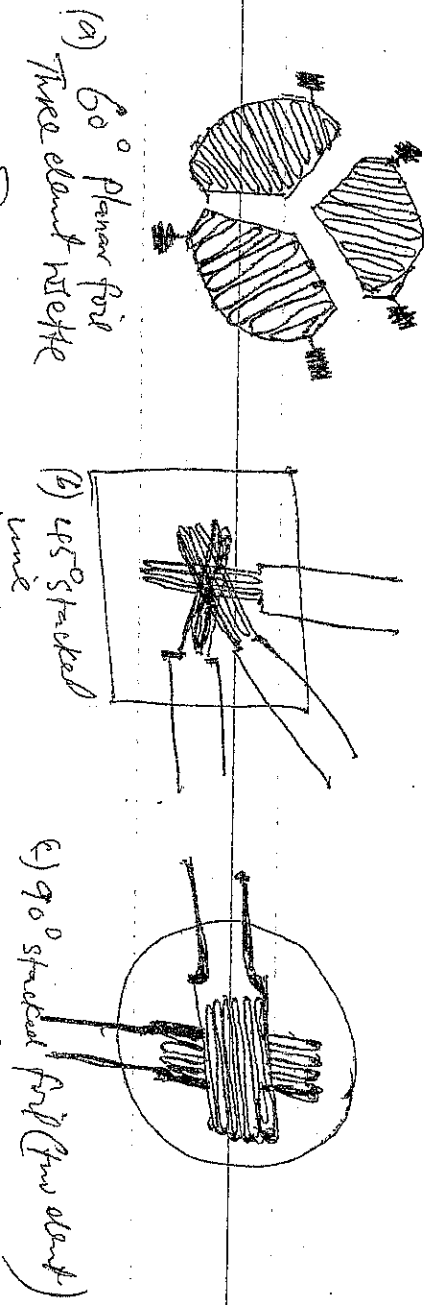


Figure Various types of rosette strain gauge

Strains that are parallel to the strain axis are measured by a single strain gauge. Whereas if there is a requirement of measurement of the strain in more than one direction, placing more than one strain gauge in a number of directions at the proper locations. Two element rosettes of four types are commonly used in force transducers. The elements are connected in a Wheatstone bridge circuit for providing maximum output. Three element rosettes are often used to determine the direction and magnitude of principal strains resulting from complex structural loading.

